

MILESTONE 4

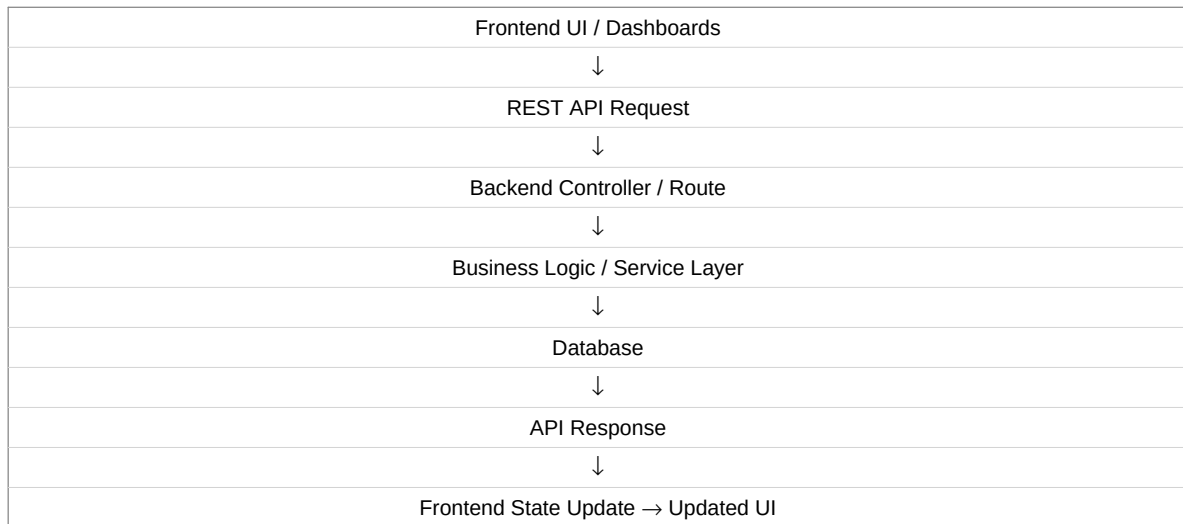
Backend Integration & End-to-End System Integration

Enterprise Learning Platform – Skill & Career Guidance System

1. Milestone Overview

Milestone 4 integrates the frontend application with backend services and the database. Earlier UI screens and dashboards are connected to actual APIs, authentication, authorization, business logic and persistent data. The objective is to establish a functional end-to-end system in which frontend actions are processed by the backend, stored in the database and reflected dynamically in the interface.

2. Technical System Architecture



Core Technical Components

- Frontend-to-backend REST API communication
- Backend routes, controllers/services, business logic and validation
- Database connectivity and persistent application data
- JWT-based authentication and protected API requests
- Role-based authorization for Student, Mentor and Admin
- Error handling, loading/empty states and data consistency

3. Integration Scope

Major integrated modules include authentication, user management, dashboards, course management, enrollment, learning progress, assignments, quizzes, sessions, achievements, certificates, notifications, complaints, approvals, audit logs and global search.

Backend & Database Integration

4. Backend API Integration

The frontend communicates with backend APIs for all major platform operations. Static or mock data is replaced by dynamic data retrieved from backend services.

API Area	Integrated Operations
Authentication	Login, registration and JWT validation
Users	Create, view, search, update, role/status management
Courses	Create, edit, draft, submit, approve/reject and publish
Enrollment & Progress	Enrollment, completion, progress percentage and learning activity
Assignments & Quizzes	Creation, submission, grading, attempts and results
Sessions	Create, update, schedule and view sessions
Achievements & Certificates	Achievement conditions, certificate records and display
Notifications	Event notifications and read/unread status
Complaints & Approvals	Complaint lifecycle and course approval workflow
Audit & Search	Activity traceability and dynamic content search

5. Database Integration

The backend is connected to the application database to provide persistent storage. Data created or modified through the application is stored and retrieved dynamically.

- Users, roles, courses and lessons
- Enrollments and learning progress
- Assignments, quizzes, attempts and results
- Achievements and certificates
- Notifications and sessions
- Complaints and audit logs

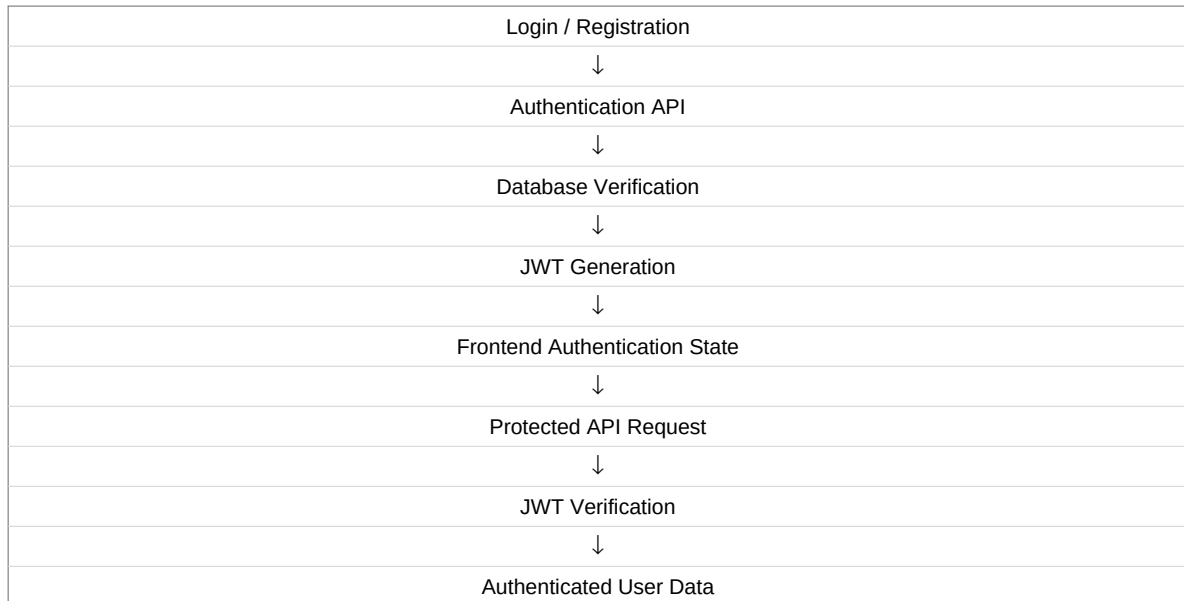
Data Persistence Flow

User Action → API Request → Validation → Business Logic → Database Write
Database Read → Backend Response → Frontend State → Updated Dashboard

Authentication, Authorization & Security

6. JWT Authentication Integration

The JWT authentication mechanism is integrated with the backend. After login or registration, the backend verifies the user, generates a JWT and the frontend maintains the authenticated state. Protected requests are verified before access is granted.



7. Role-Based Authorization

The backend determines the authenticated user's role and applies authorization rules to restricted APIs and resources.

Role	Primary Access
Student	Student dashboard, courses, learning, assignments, quizzes, achievements and certificates
Mentor	Mentor dashboard, course creation, assignments, quizzes and sessions
Admin	Admin dashboard, user management, approvals, complaints, audit and platform operations

8. Security Integration

- JWT authentication and authentication expiry
- Password hashing
- Role-based access control (RBAC)
- Protected API routes
- Input validation and permission checking
- Secure API communication and protected user data
- Secure database access

9. Error Handling & Validation

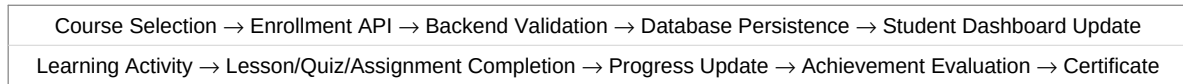
The application handles invalid requests, authentication and authorization failures, validation errors, missing resources, database errors, network/API errors, expired authentication, duplicate records and server errors. User-facing messages are provided without exposing sensitive backend information.

Core Module Integration & End-to-End Workflows

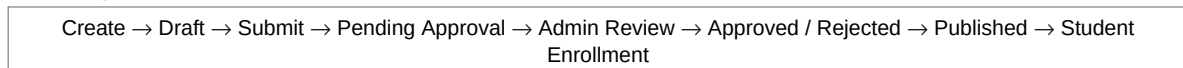
10. Dashboard Integration

- **Student Dashboard:** profile, XP, active/completed courses, study hours, streak, achievements, certificates, progress, notifications and recommendations.
- **Mentor Dashboard:** students, created/published/draft courses, pending assignments/quizzes, enrollments, activity, notifications and upcoming sessions.
- **Admin Dashboard:** users, courses, pending approvals, complaints, audit logs, notifications and platform activity.

11. Core Learning Workflow



Course Lifecycle



12. Assignment & Quiz Integration

- Students can view and submit assignments and track submission status.
- Mentors can create assignments, review submissions, grade work and provide feedback.
- Quizzes support creation, questions, attempts, submission, result calculation, score storage and student result display.

13. Supporting Modules

- Sessions: creation, scheduling, updates and student access.
- Achievements: backend evaluation of enrollment, completion, streak, quiz and skill conditions.
- Certificates: completion-based generation/storage, student association and status tracking.
- Notifications: event-driven notifications with read/unread state.
- Complaints: creation, listing, status updates, resolution and history.
- Audit logs: user, action, resource, timestamp and description for traceability.
- Search: dynamic search across courses, roadmaps, mentors and learning resources.

Testing, Deliverables & Milestone Outcome

14. Data Consistency

The system maintains consistency between frontend and backend. For example, after enrollment, the enrollment record is created, the database is updated, the student dashboard is refreshed, the active-course count changes and the course appears in the student's enrolled courses.

15. UI States

- **Loading State:** displayed while backend data is being retrieved.
- **Empty State:** displayed when no data is available.
- **Error State:** displayed when data retrieval or processing fails.

16. Technical Deliverables

Layer	Key Deliverables
Frontend	Dashboard, authentication, user, course, enrollment, progress, assignment, quiz and supporting APIs connected.
Backend	API routes, controllers/services, business logic, authentication/authorization middleware, validation and error handling.
Database	Persistent storage for users, courses, enrollments, progress, assignments, quizzes, achievements, certificates, notifications, sessions, complaints and audit data.
Security	JWT, password hashing, RBAC, protected routes, validation, authentication expiry and permission checking.

17. End-to-End System Outcome

Milestone 4 moves the Enterprise Learning Platform from interface-level functionality to a fully integrated application. Frontend, backend, database, authentication, authorization and business logic operate together as one system.

UI → API → Backend → Database → API → UI
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Final Outcome: The platform supports integrated workflows for Students, Mentors and Administrators with persistent data, protected APIs and dynamic dashboard updates.